

AUD RATES • QUANT SERIES

# AUD Swaps & Curves

Pricing, curve building and risk  
with QuantLib Python



Rates with Bip

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**Rates with Bip**

**AUD Swaps & Curves.** Pricing, curve building and risk with QuantLib Python.

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The swap, basis and BBSW quotes in this book are illustrative, made up for teaching. They are not market data. The AONIA fixings, cash rate target and Cash Rate Total Return Index in Chapters 1 and 2 are published by the Reserve Bank of Australia (Source: RBA 2026); the RBA publishes them free of charge and does not endorse this book.

Market conventions are checked against the sources listed at the end of each chapter. Conventions change; always confirm against the current documents of AFMA, the RBA, ASX and your clearing house before relying on them.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

# Contents

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|                                       |    |
|---------------------------------------|----|
| How to use this book                  | 5  |
| 1. The AUD Rates Landscape            | 7  |
| 2. How an AONIA Swap Pays             | 13 |
| 3. Bootstrapping the AONIA Curve      | 19 |
| 4. BBSW Swaps                         | 27 |
| 5. Dual-Curve Pricing                 | 33 |
| 6. Basis Swaps                        | 38 |
| 7. Valuing a Live Swap                | 44 |
| 8. Risk I: DV01                       | 49 |
| 9. Risk II: Bucketed Delta            | 53 |
| Appendix: AUD conventions at a glance | 58 |

# How to use this book

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This book follows the nine video episodes of the course, one chapter per episode, and goes a little deeper in each. It is written for people who work with Australian dollar interest rates, or want to: graduates and junior staff on rates desks, in market risk, product control and treasury, and quants who know the maths from another market but not the Australian conventions.

The path runs from one overnight rate to a hedged book:

- **Chapters 1–2** set the scene: the cash rate, AONIA and BBSW, and exactly how an AONIA swap pays, using real fixings from the Reserve Bank of Australia.
- **Chapters 3–5** build curves: the AONIA curve from overnight index swaps, the BBSW curve from FRAs and swaps, and the two-curve set-up the market actually uses.
- **Chapters 6–7** complete the toolkit: basis swaps, which tie AONIA, 3-month and 6-month BBSW into one family of curves, and valuing a swap that has been live for a year.
- **Chapters 8–9** turn to risk: DV01 in its different forms, convexity, risk by maturity bucket as the Basel FRTB rules define it, and hedging.

## Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion QuantLib Python notebook, one per chapter. Nothing is typed in by hand. Where a chapter works an example by hand, it also shows that QuantLib gets the same answer, usually to within a fraction of a cent. You are encouraged to run the notebooks, change the inputs and watch the results move.

The notebooks are free to download from [github.com/RatesWithBip/curriculum](https://github.com/RatesWithBip/curriculum), in the `aud-swaps-and-curves` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data, so there is nothing else to download.

## About the data

Most quotes in this book are **illustrative**: swap rates, FRA and basis quotes made up for teaching, shaped like a plausible Australian dollar curve but not market data. BBSW and dealer quotes are licensed data, and the ideas don't depend on the exact levels. Chapters 1 and 2 use real cash rate data published by the Reserve Bank of Australia.

## About the conventions

Market conventions are where most mistakes hide, so every convention stated in this book is checked against a primary source, mainly AFMA's *Interest Rate Derivative Conventions*, the RBA's cash rate documentation and ASX's BBSW methodology. The sources are listed at the end of each chapter. Where the RBA and AFMA differ on AONIA itself, such as the publication time, this book follows the RBA as the rate's administrator.

Conventions do change. Before relying on one in practice, check the current version of the source, and your clearing house's product terms.

# The AUD Rates Landscape

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*Three rates run the Australian dollar money market: the cash rate target, AONIA and BBSW. This chapter introduces each, shows fifteen years of real data, and sets out which derivatives use which rate.*

## Three rates

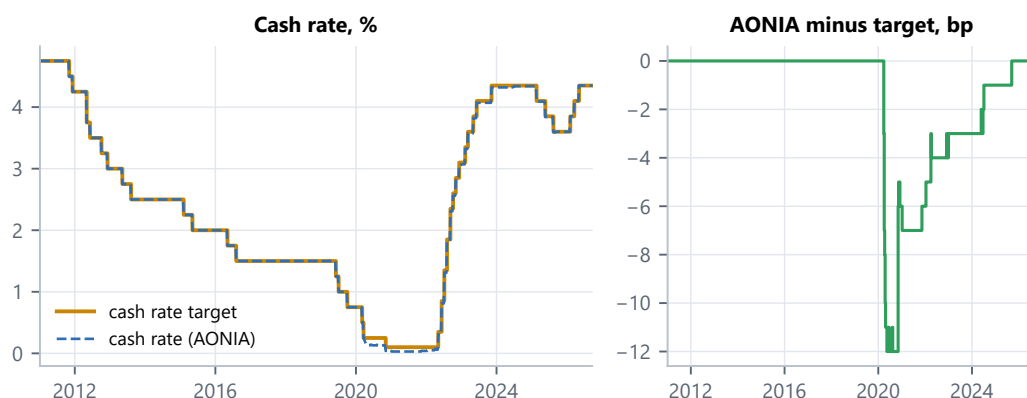
**The cash rate target** is the Reserve Bank of Australia's policy rate. The Monetary Policy Board decides it at its meetings, eight a year. Decisions are published at 2:30pm Sydney time after each meeting, and a change takes effect the following day.

**The cash rate** is the interest rate on unsecured overnight loans between banks. It is the market outcome the RBA steers towards the target, and it is calculated and published by the RBA from data banks report directly. In derivatives markets it is called **AONIA**, and it is the floating rate of Australian dollar overnight index swaps.

**BBSW**, the Bank Bill Swap Rate, is the market rate on short-term bank paper: bank bills and negotiable certificates of deposit issued by a group of prime banks, for terms of one to six months. ASX calculates and publishes it every business day. Unlike the cash rate, BBSW includes a credit premium: it is what banks pay to borrow for a term, not overnight.

## Fifteen years of the cash rate

Figure 1.1 shows the target and the cash rate from January 2011 to September 2026, using the RBA's published data. The target started the period at 4.75%. It was cut, step by step, to a low of 0.1% in November 2020. From May 2022 it rose 13 times in a row, 425 basis points in all, to 4.35% in November 2023. Since then it has been cut 3 times and raised 3 times, and stands at 4.35%. Over the whole period the target changed 37 times.



**Figure 1.1.** The cash rate target and the cash rate (AONIA), weekly, and the gap between them in basis points. Source: RBA 2026 (Statistical Table F1).

## AONIA and the target

Most of the time the cash rate sits exactly on the target. Before March 2020 the RBA supplied just enough reserves to the banking system each day for that to happen, and it almost always did: in our data the cash rate equalled the target on every day before Wednesday 25 March 2020.

From 2020 the banking system held far more reserves than it needed, and the cash rate traded a little below the target: by up to 12 basis points, first reached on Monday 27 April 2020. Between March 2020 and the end of 2025 it was on target on only a small share of days. Since Wednesday 1 October 2025 it has matched the target again.

| PERIOD            | AVERAGE GAP (BP) | WIDEST GAP (BP) | SHARE OF DAYS ON TARGET |
|-------------------|------------------|-----------------|-------------------------|
| Before March 2020 | 0.0              | 0.0             | 1.00                    |
| March 2020 – 2025 | -4.2             | -12.0           | 0.07                    |
| 2026              | 0.0              | 0.0             | 1.00                    |

For a swap, this matters in one direction only: an AONIA swap pays the cash rate actually published each day, not the target. When the two differ, the swap follows the cash rate.

## How BBSW is set

ASX administers BBSW under its published conventions and methodology. Each business day:

1. Prime banks' bills and certificates of deposit trade in the market. Trades between **8:30 and 10:00am** Sydney time are eligible.
2. For each tenor from one to six months, ASX calculates a **volume-weighted average** of the eligible trades (or a weighted least-squares fit where trades mature on different dates).
3. If there aren't enough trades in a tenor, ASX falls back to the **national best bid and offer**, and beyond that to a fall-back process.
4. The rates are rounded to four decimal places and published by **10:30am**.

ASX describes BBSW as a rate that includes a credit premium: the extra the prime banks pay over a comparable risk-free curve.

## AONIA and BBSW compared

The two rates answer different questions, and it shows in how derivatives use them.

|               | AONIA                                        | BBSW                                |
|---------------|----------------------------------------------|-------------------------------------|
| Term          | overnight                                    | one to six months                   |
| Credit        | near risk-free                               | includes a bank credit premium      |
| For a period  | compounded day by day, known only at the end | forward-looking, known at the start |
| Administrator | Reserve Bank of Australia                    | ASX                                 |

Because BBSW reflects the cost of lending to banks for months rather than overnight, it has usually been slightly higher than AONIA. The RBA's own description of the BBSW fallback makes the same point: if BBSW ever stopped, contracts would switch to AONIA **compounded in arrears plus a spread**, where compounding matches the tenor and the spread covers the credit risk.

## Why Australia kept BBSW

When LIBOR was retired overseas, regulators pushed markets to move wholesale onto risk-free rates. Australia took a different path. BBSW's methodology and infrastructure had been strengthened, so it stayed, alongside AONIA. The RBA calls this a **multiple-rate approach**: rather than a wholesale transition, market participants choose the reference rate that best suits each product. Floating rate notes issued by governments, companies and securitisation trusts might reference the cash rate, while notes and loans issued by banks can keep using BBSW.

## Which product uses which rate

In derivatives, most Australian dollar interest rate swaps settle against BBSW; overnight index swaps settle against the compounded cash rate.

Standard AUD interest rate derivatives and their floating rates (AFMA).

| PRODUCT                       | FLOATING RATE             |
|-------------------------------|---------------------------|
| AONIA OIS                     | AONIA, compounded daily   |
| BBSW swap, up to 3 years      | 3-month BBSW, quarterly   |
| BBSW swap, 4 years and longer | 6-month BBSW, semi-annual |
| 3s6s basis swap               | 3-month vs 6-month BBSW   |
| Cash/BBSW basis swap          | AONIA vs BBSW             |
| FRA                           | BBSW for the period       |

Vanilla swaps pay quarterly against 3-month BBSW out to three years, and semi-annually against 6-month BBSW from four years; swaps between three and four years are negotiated. Basis swaps exchange one floating rate for another, and are the subject of Chapter 6.

## Dates: T+1 and Modified Following

Dates follow conventions too, and AFMA's conventions work through an example.

- An Australian dollar swap **starts on the next Sydney business day** after the trade: T+1.
- Later dates keep the same day of the month and roll **Modified Following**: a date on a weekend or holiday moves forward to the next business day, unless that crosses into the next month, in which case it moves back instead.
- Single-currency AUD swaps have **no end-of-month rule**. (Cross-currency swaps do, to match USD conventions.)

A swap traded on Monday 29 April 2024 starts on Tuesday 30 April 2024. The table shows the first three monthly dates.

AFMA §3.3 worked example, reproduced in QuantLib.

| TENOR | SAME DAY    | WEEKDAY  | AUD SWAP    | CROSS-CURRENCY (EOM) |
|-------|-------------|----------|-------------|----------------------|
| 1M    | 30 May 2024 | Thursday | 30 May 2024 | 31 May 2024          |
| 2M    | 30 Jun 2024 | Sunday   | 28 Jun 2024 | 28 Jun 2024          |
| 3M    | 30 Jul 2024 | Tuesday  | 30 Jul 2024 | 31 Jul 2024          |

The two-month date would be Sunday 30 June. Rolling forward would cross into July, so it rolls back to Friday 28 June 2024. And although the swap starts on the last business day of April, the one-month date stays on 30 May: with no end-of-month rule, it does not jump to 31 May as the cross-currency date does.

Forward-starting swaps count from the trade's spot date. AFMA's example is a one-year swap starting in three months, traded on Thursday 20 June 2024: spot is Friday 21 June 2024, the start date Monday 23 September 2024 (the 21st is a Saturday) and the end date Monday 22 September 2025 (the 21st is a Sunday).

Reproducing AFMA's date examples with QuantLib's Australian calendar.

```
cal = ql.Australia(ql.Australia.Settlement)
# QuantLib 1.43 misses NSW's additional Anzac Day holidays when 25 April falls on a weekend.
for d in [ql.Date(27, 4, 2026), ql.Date(26, 4, 2027)]:
    cal.addHoliday(d)
sydney_ny = ql.JointCalendar(cal, ql.UnitedStates(ql.UnitedStates.FederalReserve))

def forward_dates(calendar, spot, months, eom):
    return [calendar.advance(spot, ql.Period(m, ql.Months), ql.ModifiedFollowing, eom) for m
            in months]

# AFMA example 1: AUD swap traded Monday 29 April 2024
trade1 = ql.Date(29, 4, 2024)
spot1 = cal.advance(trade1, 1, ql.Days)
aud = forward_dates(cal, spot1, [1, 2, 3], eom=False)
# AFMA example 2: cross-currency swap traded Friday 26 April 2024, spot T+2, end-of-month rule
trade2 = ql.Date(26, 4, 2024)
spot2 = sydney_ny.advance(trade2, 2, ql.Days)
xccy = forward_dates(sydney_ny, spot2, [1, 2, 3], eom=True)
# AFMA example 3: one-year swap starting in three months (3m/1y), traded 20 June 2024
trade3 = ql.Date(20, 6, 2024)
spot3 = cal.advance(trade3, 1, ql.Days)
start3 = cal.adjust(spot3 + ql.Period(3, ql.Months), ql.ModifiedFollowing)
end3 = cal.adjust(spot3 + ql.Period(15, ql.Months), ql.ModifiedFollowing)

AFMA_PRINTED = { # dates as printed in AFMA §3.3
```

**Check it yourself.** The notebook compares every date QuantLib produces with the dates printed in AFMA §3.3. Do they all match? **yes.**

## Summary

- The RBA sets the **cash rate target**; the **cash rate**, known in derivatives as **AONIA**, is where overnight money actually trades, and it is what AONIA swaps pay.
- **BBSW** is a term rate with a bank credit premium, set by ASX from trades each morning.

- Australia kept both rates: a **multiple-rate approach**.
- Most AUD swaps pay BBSW: quarterly against 3-month out to three years, semi-annual against 6-month from four.
- Swaps start at **T+1** and roll **Modified Following**, with **no end-of-month rule**.

---

## Sources

Companion notebook: `ep01_aud_rates_landscape.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
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3. Reserve Bank of Australia, *Fallbacks for BBSW Securities (Bulletin, June 2022)*. <https://www.rba.gov.au/publications/bulletin/2022/jun/fallbacks-for-bbsw-securities.html>
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9. Reserve Bank of Australia, *Benchmark Reforms (speech by Christopher Kent, Assistant Governor (Financial Markets), 17 November 2020)*. <https://www.rba.gov.au/speeches/2020/sp-ag-2020-11-17.html>
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# How an AONIA Swap Pays

*We follow one three-month AONIA swap from trade to settlement, using the cash rates the RBA actually published in early 2026, and check the payment three independent ways.*

## The trade

An **overnight index swap** (OIS) exchanges a fixed rate for a floating rate that is built up from an overnight index, day by day. In Australian dollars the index is AONIA. Our example is a three-month swap in which we receive fixed.

| TERM                      | VALUE                    |
|---------------------------|--------------------------|
| Side                      | receive fixed, pay AONIA |
| Notional                  | AUD 100 million          |
| Fixed rate (illustrative) | 3.80%                    |
| Traded                    | Monday 19 January 2026   |
| Start (T+1)               | Tuesday 20 January 2026  |
| End                       | Monday 20 April 2026     |
| Payment                   | Wednesday 22 April 2026  |

The fixed rate is made up for teaching, but every AONIA fixing in this chapter is real: the cash rate the RBA published for each day of the swap (Source: RBA 2026).

The dates follow the AFMA conventions from Chapter 1. The swap starts on the next Sydney business day after the trade and runs 90 days to its end date. Because it is no longer than twelve months, there is a **single payment**, at maturity. And an AONIA swap pays **two business days after the end date**, not on it; we come back to the timing at the end of the chapter.

## One fixing per business day

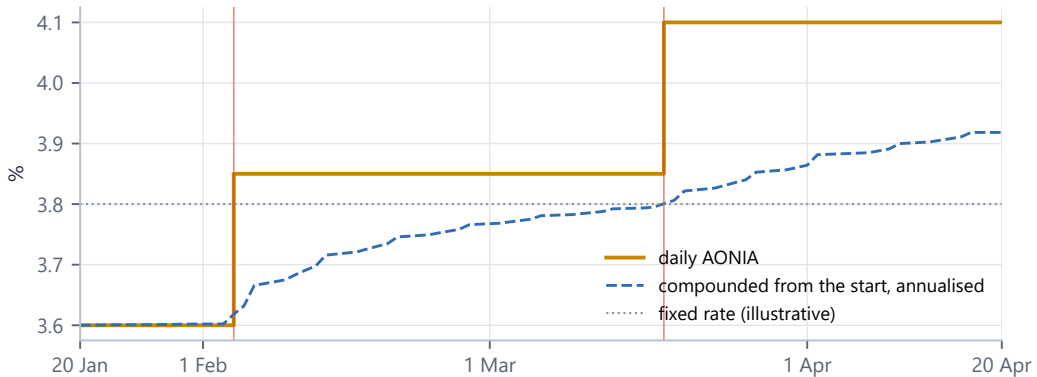
The floating leg uses one AONIA fixing for each Sydney business day from the start date up to, but not including, the end date: 61 fixings here, covering 90 calendar days.

Each fixing accrues until the next business day. On an ordinary weekday that is one day. A Friday's rate covers the weekend, so it counts three days. AFMA's conventions spell this out: non-business days are included as extra days in the day count of the previous business day. Holidays stretch it further. The Friday before Australia Day on Monday 26 January counted 4 days, and over Easter the Thursday's rate covered 5 days: Good Friday, the weekend and Easter Monday.

A selection of the fixings: the first and last days, the long weekends, and the days either side of each rate change. Source: RBA 2026.

| DAY        | AONIA % | DAYS | $1 + R \cdot N/365$ |
|------------|---------|------|---------------------|
| Tue 20 Jan | 3.60    | 1    | 1.00009863          |
| Fri 23 Jan | 3.60    | 4    | 1.00039452          |
| Tue 3 Feb  | 3.60    | 1    | 1.00009863          |
| Wed 4 Feb  | 3.85    | 1    | 1.00010548          |
| Tue 17 Mar | 3.85    | 1    | 1.00010548          |
| Wed 18 Mar | 4.10    | 1    | 1.00011233          |
| Thu 2 Apr  | 4.10    | 5    | 1.00056164          |
| Fri 17 Apr | 4.10    | 3    | 1.00033699          |

The rate itself moved during the swap. The Reserve Bank raised the cash rate twice: from 3.60% to 3.85% (from Wednesday 4 February 2026) and then to 4.10% (from Wednesday 18 March 2026).



**Figure 2.1.** The daily AONIA fixing over the life of the swap (solid), the rate compounded from the start date to each day, annualised (dashed), and our fixed rate (dotted). Red lines mark the two rate rises. Source: RBA 2026.

The dashed line lags the solid one: every day at the old, lower rate stays in the compounded average.

## The settlement formula

With  $N$  the notional,  $S$  the fixed rate and  $D$  the number of days from start to end, AFMA's formulas are

$$\text{Fixed} = N \cdot S \cdot \frac{D}{365}$$

$$\text{Float} = N \left[ \prod_{i=1}^n \left( 1 + r_i \frac{n_i}{365} \right) - 1 \right]$$

where  $r_i$  is the fixing for business day  $i$  and  $n_i$  the number of days it covers. Both legs count days as actual/365 fixed. The two amounts are netted, the floating amount rounded to the nearest cent, and whoever owes pays the difference.

## Compounding by hand

Multiplying the 61 daily factors together gives a growth factor of 1.0096617345. So

- the floating amount is AUD 966,173.45, and
- the fixed amount is AUD 100 million  $\times$  3.80%  $\times$  90/365 = AUD 936,986.30.

Expressed as a simple rate over the 90 days, the floating leg paid 3.9184%.

## Compounding versus averaging

Why not just average the fixings? The day-weighted average of the rates is 3.9000%. The compounded rate is 3.9184%, about 1.8 basis points higher. The reason is that the product of the daily factors is always a little more than one plus the sum of the daily interest:

$$\prod_i (1 + x_i) = 1 + \sum_i x_i + \sum_{i < j} x_i x_j + \dots \geq 1 + \sum_i x_i$$

The cross terms are interest earned on interest. They are small, but on AUD 100 million over three months they are worth about AUD 4,500. A swap that averaged instead of compounding would be a different product.

## Three ways to the same payment

We can check the floating amount three independent ways.

1. **By hand**, as above.
2. **With QuantLib**: load the RBA fixings into an AONIA index, build an overnight indexed swap with the same schedule and a two-day payment lag, and set the evaluation date to the end of the swap so every rate is a past fixing and nothing is forecast.
3. **With the RBA's own index**. The RBA publishes a Cash Rate Total Return Index that reinvests the cash rate on every business day, each rate accruing until the next business day:

$$\text{TRI}_t = \text{TRI}_{t-d} \left( 1 + \text{CashRate}_{t-d} \frac{d}{365 \times 100} \right)$$

That is exactly the compounding of the OIS floating leg, so the ratio of the index on the end and start dates should give the same growth factor.

Historical fixings in QuantLib: add the fixings, then build and price the swap as of its end date.

```
for r in rba.itertuples():
    d = ql.Date(r.date.day, r.date.month, r.date.year)
    if r.date.year >= 2025 and cal.isBusinessDay(d):
        aonia.addFixing(d, r.cash_rate / 100)

ql.Settings.instance().evaluationDate = end
schedule = ql.Schedule(start, end, ql.Period(ql.Once), cal, ql.ModifiedFollowing,
ql.ModifiedFollowing,
                        ql.DateGeneration.Backward, False)
swap = ql.OvernightIndexedSwap(ql.Swap.Receiver, notional, schedule, S, dc, aonia, 0.0,
                               2, ql.Following, cal) # payment lag: two business days
fixed_cf, float_cf = swap.fixedLeg()[0], swap.overnightLeg()[0]
coupon = ql.as_overnight_indexed_coupon(float_cf)
qlres = {
    "float_amount": float_cf.amount(), "fixed_amount": fixed_cf.amount(),
    "compounded_rate_pct": coupon.rate() * 100, "payment_date": iso(float_cf.date()),
    "n_fixings": len(coupon.fixingDates()),
}
qlres["abs_diff_float"] = abs(qlres["float_amount"] - hand["float_amount"])
out["quantlib"] = qlres
pd.Series(qlres)
```

|                 | BY HAND        | QUANTLIB       | RBA INDEX      |
|-----------------|----------------|----------------|----------------|
| Floating amount | AUD 966,173.45 | AUD 966,173.45 | AUD 966,174.10 |
| Compounded rate | 3.9184%        | 3.9184%        | 3.9184%        |

The hand calculation and QuantLib agree to the cent, using 61 fixings. The RBA index agrees to within 66 cents: the index is rounded to six decimal places each day, and those roundings add up to a few cents on this notional.

**Check it yourself.** The index values on the start and end dates are 142.431990 and 143.808131. Their ratio minus one, times the notional, is the floating amount. This is a quick, free way to check the floating leg of any AONIA swap.

## Who pays?

We receive AUD 936,986.30 and owe AUD 966,173.45. The two rate rises pushed AONIA above our fixed rate, so the net payment of AUD 29,187.15 runs from us, the fixed receiver, on Wednesday 22 April 2026.

The fixed rate that would have broken even is the compounded rate itself, 3.9184%. When the swap was traded, nobody knew what that would be; that uncertainty is exactly what the swap exchanges.

## When the payment is known

The timing is worth following through to the end.

- Each day's cash rate is published by the RBA on the **next** business day, before 9:20am.
- The last fixing is for the business day before the end date, here Friday 17 April 2026, and it is published on the end date itself, Monday 20 April 2026.
- So the floating amount is known only on the end date, and it is paid two business days later.

## Summary

- An AONIA swap compounds **one fixing per business day**, each weighted by the days it covers until the next business day.
- **Compounding** adds a little over simple averaging: interest on interest.
- The hand calculation, QuantLib and the RBA's **Cash Rate Total Return Index** all give the same floating amount.
- The amount is known on the end date and **paid two business days later**.

## Sources

Companion notebook: `ep02_aonia_compounding.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Reserve Bank of Australia, *Cash Rate Methodology - Overview*. <https://www.rba.gov.au/mkt-operations/resources/cash-rate-methodology/>
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# Bootstrapping the AONIA Curve

*Almost every Australian dollar swap is discounted on one curve, the AONIA curve. We build it from overnight index swap quotes, check it by hand, look at what interpolation does, and measure a first DV01.*

## Why this curve matters

AFMA's conventions require prices for Australian dollar derivatives to be quoted as if their cash flows are discounted in the same way as cleared derivatives, which for AUD swaps currently means discounting on **AUD OIS curves**, whether or not the trade is cleared. So the AONIA curve is the foundation for everything that follows: it turns future cash flows into today's values.

## AONIA swap conventions

Chapter 2 showed how one period of an AONIA swap pays. Longer swaps add a few more rules.

| ITEM            | CONVENTION                                                           |
|-----------------|----------------------------------------------------------------------|
| Start           | next Sydney business day (T+1)                                       |
| Date rolling    | Modified Following, no end-of-month rule                             |
| Up to 12 months | one payment, at maturity                                             |
| Over 12 months  | a front stub if needed, then annual payments                         |
| Payment         | two business days after each period ends                             |
| Day count       | actual/365 fixed, both legs                                          |
| Compounding     | daily; weekends and holidays count towards the previous business day |
| Last fixing     | the business day before the end date                                 |

For one period from  $s$  to  $e$ , with fixed rate  $S$  and  $D$  days, the legs are

$$\text{Fixed} = N \cdot S \cdot \frac{D}{365}, \quad \text{Float} = N \left[ \prod_{i=1}^n \left( 1 + r_i \frac{d_i}{365} \right) - 1 \right]$$

## Our quotes

Illustrative quotes: an overnight deposit and AONIA swaps out to ten years. Not market data.

| TENOR | INSTRUMENT | RATE (%) |
|-------|------------|----------|
| O/N   | DEPOSIT    | 3.85     |
| 1M    | OIS        | 3.86     |
| 3M    | OIS        | 3.89     |
| 6M    | OIS        | 3.93     |
| 9M    | OIS        | 3.97     |
| 1Y    | OIS        | 4.00     |
| 18M   | OIS        | 4.05     |
| 2Y    | OIS        | 4.08     |
| 3Y    | OIS        | 4.13     |
| 5Y    | OIS        | 4.24     |
| 7Y    | OIS        | 4.35     |
| 10Y   | OIS        | 4.50     |

The overnight point needs a stand-in: today's AONIA fixing isn't published until tomorrow morning, so we use an overnight rate of 3.85% for the first night.

## The bootstrap idea

Bootstrapping solves for discount factors one maturity at a time, each quote pinning down the next one.

A swap of up to a year has a single period, from start  $s$  to end  $e$ , paid at  $p$ . If the same curve both forecasts AONIA and discounts, the compounded floating rate for the period is exactly the ratio of discount factors, so

$$PV_{\text{float}} = \left( \frac{P(s)}{P(e)} - 1 \right) P(p), \quad PV_{\text{fixed}} = S \tau P(p)$$

with  $\tau = D/365$ . Setting the two equal, the payment-date factor  $P(p)$  cancels:

$$P(e) = \frac{P(s)}{1 + S \tau}$$

Longer swaps have several periods. Without a payment lag the floating leg telescopes, each period's end factor cancelling the next period's start, and the par rate is

$$S = \frac{P(t_0) - P(t_n)}{\sum_i \tau_i P(t_i)}$$

which can be solved for the last unknown factor  $P(t_n)$ , given all the earlier ones. With the two-day payment lag the cancellation is no longer exact, and QuantLib solves the exact condition numerically instead.

## Building the curve in QuantLib

In QuantLib each quote becomes a **rate helper**, and the conventions map straight onto its arguments: one settlement day, a payment lag of two, annual payments, Modified Following and no end-of-month rule. A piecewise log-linear discount curve then bootstraps all the helpers together.

Quotes to rate helpers, and the bootstrap.

```
def make_helpers():
    # Fresh helpers and quotes each call: a helper can only belong to one curve.
    qs, hs = {}, []
    for row in quotes.itertuples():
        q = ql.SimpleQuote(row.rate_pct / 100)
        qs[row.tenor] = q
        if row.instrument == "deposit":
            h = ql.DepositRateHelper(ql.QuoteHandle(q), ql.Period(1, ql.Days), 0, cal,
                                     ql.Following, False, dc)
        else:
            h = ql.OISRateHelper(CONV["settlement_days"], ql.Period(row.tenor),
                                ql.QuoteHandle(q), aonia,
                                paymentLag=CONV["payment_lag_days"],
                                paymentFrequency=ql.Annual,
                                paymentCalendar=cal, convention=ql.ModifiedFollowing,
                                endOfMonth=False)
            hs.append((row.tenor, row.instrument, h))
    return qs, hs

quote_handles, helpers = make_helpers()
curve = ql.PiecewiseLogLinearDiscount(today, [h for _, h in helpers], dc)
curve.enableExtrapolation()
curve_handle = ql.YieldTermStructureHandle(curve)

pillars = pd.DataFrame([
    "tenor": tenor,
    "instrument": kind,
    "quote_pct": quote_handles[tenor].value() * 100,
    "maturity": iso(h.maturityDate()),
```

The bootstrapped curve at each pillar.

| TENOR | PILLAR DATE | YEARS   | DISCOUNT FACTOR | ZERO RATE (% , CONT. ) |
|-------|-------------|---------|-----------------|------------------------|
| O/N   | 23 Sep 2026 | 0.0027  | 0.99989453      | 3.8498                 |
| 1M    | 27 Oct 2026 | 0.0959  | 0.99631142      | 3.8538                 |
| 3M    | 29 Dec 2026 | 0.2685  | 0.98965867      | 3.8717                 |
| 6M    | 25 Mar 2027 | 0.5041  | 0.98057019      | 3.8922                 |
| 9M    | 25 Jun 2027 | 0.7562  | 0.97085014      | 3.9123                 |
| 1Y    | 27 Sep 2027 | 1.0137  | 0.96102081      | 3.9222                 |
| 18M   | 27 Mar 2028 | 1.5123  | 0.94151927      | 3.9846                 |
| 2Y    | 27 Sep 2028 | 2.0164  | 0.92250742      | 4.0001                 |
| 3Y    | 26 Sep 2029 | 3.0137  | 0.88510204      | 4.0499                 |
| 5Y    | 25 Sep 2031 | 5.0110  | 0.81177883      | 4.1614                 |
| 7Y    | 27 Sep 2033 | 7.0192  | 0.74065659      | 4.2771                 |
| 10Y   | 25 Sep 2036 | 10.0164 | 0.64094632      | 4.4408                 |

## Checking it by hand

Before trusting the library, check the first two pillars by hand.

With the overnight rate  $r_{O/N} = 3.85\%$  over one night, the spot discount factor is

$$P(\text{spot}) = \frac{1}{1 + r_{O/N} \cdot 1/365} = 0.9998945317$$

The one-month swap, quoted at  $S_{1M} = 3.86\%$ , runs 30 days from Wednesday 23 September 2026 to Friday 23 October 2026, so

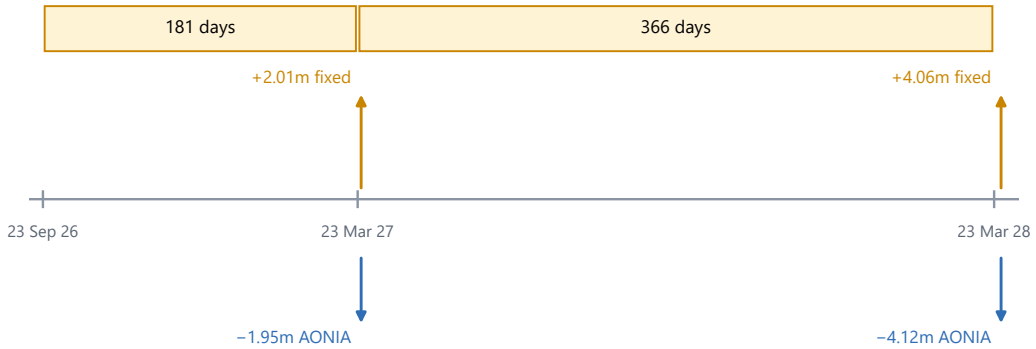
$$P(1M) = \frac{P(\text{spot})}{1 + S_{1M} \cdot 30/365} = 0.9967322961$$

|                     | BY HAND      | QUANTLIB     |
|---------------------|--------------|--------------|
| $P(\text{spot})$    | 0.9998945317 | 0.9998945317 |
| $P(1M \text{ end})$ | 0.9967322961 | 0.9967322961 |

QuantLib agrees to within one ten-trillionth. If the two ever disagree, the conventions in the code are wrong.

## Cash flows of an 18-month swap

An 18-month swap is longer than a year, so it starts with a **front stub** and then pays annually.



Receive fixed, AUD 100m. Each payment is made two business days after its period ends.

**Figure 3.1.** An 18-month AONIA swap at par, receiving fixed on AUD 100 million: a front stub of 181 days, then a full year.

The stub runs 181 days. The second period runs 366 days: it includes 29 February 2028, and actual/365 fixed does not adjust for leap years. Each payment lands two business days after its period ends.

Notice the stub's floating rate: 3.93%. That is exactly the six-month quote of 3.93%, because the stub *is* the six-month swap. At the par rate of 4.05%, the two legs have the same value.

## Does the curve reprice its inputs?

A bootstrapped curve must reprice every instrument it was built from.

Each quote against the rate implied by the curve.

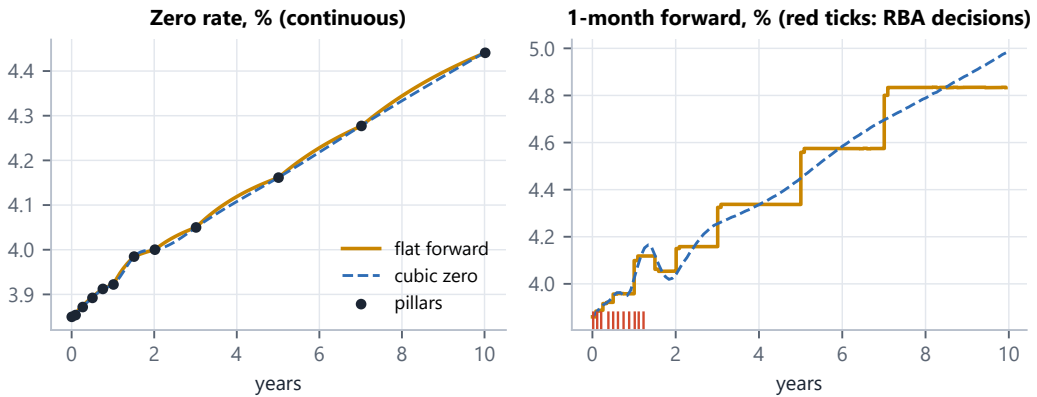
| TENOR | QUOTE (%) | CURVE (%) | ERROR (BP) |
|-------|-----------|-----------|------------|
| O/N   | 3.8500    | 3.8500    | 6.3e-10    |
| 1M    | 3.8600    | 3.8600    | 9.4e-09    |
| 3M    | 3.8900    | 3.8900    | -1.0e-11   |
| 6M    | 3.9300    | 3.9300    | -1.8e-10   |
| 9M    | 3.9700    | 3.9700    | -4.4e-10   |
| 1Y    | 4.0000    | 4.0000    | -9.5e-11   |
| 18M   | 4.0500    | 4.0500    | -1.2e-12   |
| 2Y    | 4.0800    | 4.0800    | -2.4e-10   |
| 3Y    | 4.1300    | 4.1300    | -4.2e-13   |
| 5Y    | 4.2400    | 4.2400    | 3.5e-13    |
| 7Y    | 4.3500    | 4.3500    | 1.4e-13    |
| 10Y   | 4.5000    | 4.5000    | -2.1e-13   |

The largest error is below one hundred-millionth of a basis point: numerical noise. If this check ever fails, either the bootstrap did not converge or two of the quotes contradict each other.

## Interpolation: same fit, different forwards

Between pillars the curve has to be interpolated, and the choice matters more than it first appears. Figure 3.2 compares two methods.

- **Log-linear discount factors** give piecewise-flat instantaneous forwards: forward rates that step at each quote's maturity.
- **A natural cubic spline on zero rates** is smoother, but can wobble between pillars.



**Figure 3.2.** Left: zero rates from the two interpolations, with the pillars. Right: one-month forward rates; red ticks mark the RBA's scheduled policy decisions.

Both curves reprice every quote exactly: the cubic curve's largest error is also below one hundred-millionth of a basis point. They agree at the pillars and differ in between, and it is the forwards that show it. Neither steps at the Reserve Bank's meetings, which is where policy actually changes. The Monetary Policy Board meets eight times a year, and desks quote swaps dated to those meetings to capture the steps; AFMA's conventions describe these as "RBA dates" swaps.

## A first DV01

Finally, a first risk number. We receive fixed on a five-year swap at the par rate of 4.24%, on AUD 100 million. We move **every** quote up and down by one basis point, rebuild the curve each time, and reprice:

$$\text{DV01} = \frac{V(-1\text{bp}) - V(+1\text{bp})}{2}$$

Par DV01 by bumping every quote, rebuilding and repricing.

```
par = quote_handles[dv01_tenor].value()
swap = make_ois(dv01_tenor, par, receive_fixed=True)

def npv_after_shift(bp):
    for q in quote_handles.values():
        q.setValue(q.value() + bp / 1e4)
    v = swap.NPV()
    for q in quote_handles.values():
        q.setValue(q.value() - bp / 1e4)
    return v

base = swap.NPV()
up, down = npv_after_shift(+1), npv_after_shift(-1)
```

| SHIFT           | VALUE             |
|-----------------|-------------------|
| all quotes +1bp | AUD -44,297       |
| all quotes -1bp | AUD 44,322        |
| <b>DV01</b>     | <b>AUD 44,309</b> |

The swap gains about AUD 44,300 for each basis point that rates fall: AUD 443 per million of notional. For scale, AFMA's customary OIS parcel carries about AUD 25,000 of risk per basis point, which for a five-year swap means a notional of about AUD 56.4 million. Chapter 8 looks at DV01 more closely.

## Summary

- AUD swaps are discounted on the **AONIA curve**, built from overnight index swaps.
- Longer AONIA swaps have a **front stub, annual payments and a two-day payment lag**.
- **Bootstrapping** solves for discount factors one quote at a time; check the first pillars by hand, and check that the curve **reprices every quote**.
- **Interpolation** doesn't change the fit, but it changes the forwards.
- **Par DV01**: bump every quote, rebuild, reprice.

## Sources

Companion notebook: `ep03_aonia_bootstrap.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Reserve Bank of Australia, *Cash Rate Methodology - Overview*. <https://www.rba.gov.au/mkt-operations/resources/cash-rate-methodology/>
3. Reserve Bank of Australia, *Board Meeting Schedules*. <https://www.rba.gov.au/schedules-events/board-meeting-schedules.html>

# BBSW Swaps

*Most Australian dollar swaps pay BBSW, not AONIA. This chapter covers the conventions of a vanilla AUD swap, builds a 3-month BBSW curve, derives the par swap rate by hand, and shows what the annuity tells you about value and risk.*

## Conventions

A vanilla Australian dollar interest rate swap exchanges a fixed rate for BBSW. Its conventions differ from an AONIA swap in two ways that matter: the floating rate is a term rate, and payments are made more often.

Standard AUD swap conventions (AFMA).

| ITEM            | CONVENTION                      |
|-----------------|---------------------------------|
| Floating rate   | 3M BBSW (quarterly), to 3 years |
| From 4 years    | 6M BBSW, semi-annual            |
| Fixing          | set at the start of each period |
| Payment         | at the end of each period       |
| Day count       | ACT/365 Fixed, both legs        |
| Start and dates | T+1, Modified Following, no EOM |

Swaps are quoted quarterly, against 3-month BBSW, out to three years, and semi-annually, against 6-month BBSW, from four years; maturities between three and four years are agreed between the two parties. Both legs count days as actual/365 fixed. As with AONIA swaps, the swap starts on the next business day and dates roll Modified Following with no end-of-month rule.

## The BBSW leg and the AONIA leg

The floating legs of the two kinds of swap behave quite differently.

|                   | BBSW LEG                           | AONIA LEG                       |
|-------------------|------------------------------------|---------------------------------|
| Rate for a period | set on the first day of the period | compounded every business day   |
| Known             | at the start of the period         | only at the end                 |
| Paid              | at the end of the period           | two business days after the end |

Each quarter, the BBSW rate is fixed on the first day of the period, so both sides know the next payment three months ahead; interest is then paid at the end of the period, in arrears.

## Quotes

To price BBSW swaps we need a curve of **3-month BBSW forward rates**. We build it from three kinds of instrument:

- today's **3-month BBSW fixing**;
- **forward rate agreements** (FRAs), each locking in 3-month BBSW for a future quarter. An FRA "3×6" covers the three months starting three months from now; AUD FRAs count days as actual/365 and settle against BBSW;
- **swaps** at two and three years, quarterly against 3-month BBSW.

Illustrative BBSW, FRA and swap quotes. Not market data.

| TENOR | INSTRUMENT | RATE (%) |
|-------|------------|----------|
| 3M    | BBSW       | 4.02     |
| 3x6   | FRA        | 4.08     |
| 6x9   | FRA        | 4.13     |
| 9x12  | FRA        | 4.17     |
| 12x15 | FRA        | 4.20     |
| 15x18 | FRA        | 4.23     |
| 18x21 | FRA        | 4.25     |
| 2Y    | Swap       | 4.17     |
| 3Y    | Swap       | 4.22     |

## The BBSW curve

Bootstrapping works just as it did for AONIA. In QuantLib the fixing becomes a deposit helper, each FRA an FRA helper and each swap a swap helper, all on the same 3-month BBSW index. One log-linear discount curve is solved to fit them all.

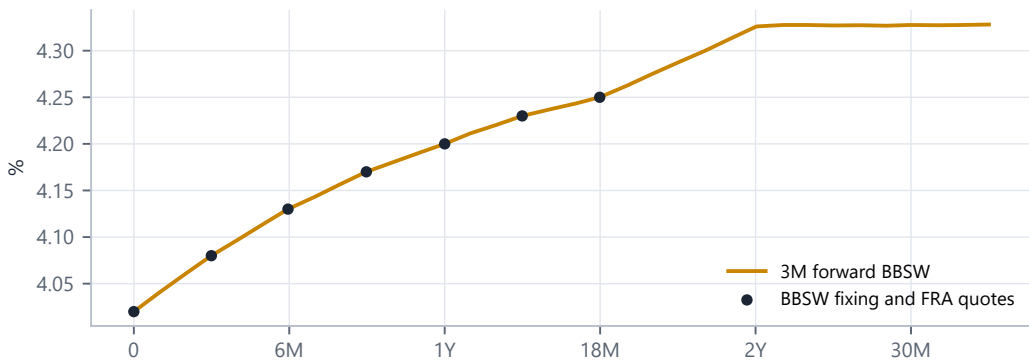
BBSW, FRA and swap quotes to rate helpers.

```
def make_helpers(index):
    qs, hs = {}, []
    for row in quotes.itertuples():
        q = ql.SimpleQuote(row.rate_pct / 100)
        qs[row.tenor] = q
        if row.instrument == "BBSW":
            h = ql.DepositRateHelper(ql.QuoteHandle(q), index)
        elif row.instrument == "FRA":
            months_to_start = int(row.tenor.split("x")[0])
            h = ql.FraRateHelper(ql.QuoteHandle(q), months_to_start, index)
        else: # quarterly swap against 3M BBSW, starting T+1
            h = ql.SwapRateHelper(ql.QuoteHandle(q), ql.Period(row.tenor), cal, ql.Quarterly,
                                  ql.ModifiedFollowing, dc, index, ql.QuoteHandle(),
                                  ql.Period(0, ql.Days),
                                  ql.YieldTermStructureHandle(), 1)
            hs.append((row.tenor, row.instrument, h))
    return qs, hs

quote_handles, helpers = make_helpers(bbsw3m())
curve = ql.PiecewiseLogLinearDiscount(today, [h for _, h in helpers], dc)
curve.enableExtrapolation()
curve_handle = ql.YieldTermStructureHandle(curve)
curve.nodes() # bootstrap now
```

In this chapter **one curve both forecasts BBSW and discounts** the cash flows. That is a deliberate simplification: in practice AUD swaps are discounted on the AONIA curve. Chapter 5 removes it and shows what changes.

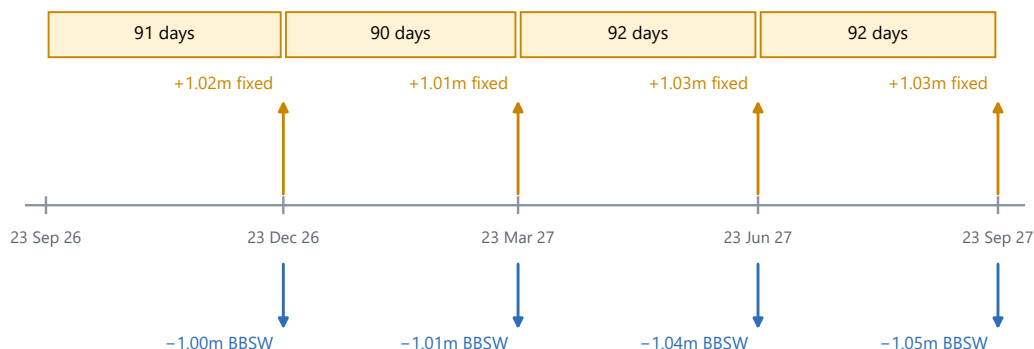
The curve reprices every input to within one hundred-millionth of a basis point. Figure 4.1 shows the 3-month forward rates it implies, passing through each FRA quote.



**Figure 4.1.** Three-month forward BBSW rates implied by the curve, with the fixing and FRA quotes at their start dates. Illustrative quotes.

## Cash flows of a one-year swap

Consider a one-year swap receiving the par rate of 4.1002% on AUD 100 million. It has four quarterly periods.



Receive fixed, AUD 100m. Each BBSW rate is set at the start of its period and paid at the end.

**Figure 4.2.** A one-year BBSW swap at par, receiving fixed on AUD 100 million.

The four quarterly periods. Each BBSW rate is the forward rate for its period.

| START       | END         | DAYS | BBSW (%) | FIXED         | FLOATING      |
|-------------|-------------|------|----------|---------------|---------------|
| 23 Sep 2026 | 23 Dec 2026 | 91   | 4.0207   | AUD 1,022,237 | AUD 1,002,412 |
| 23 Dec 2026 | 23 Mar 2027 | 90   | 4.0806   | AUD 1,011,003 | AUD 1,006,163 |
| 23 Mar 2027 | 23 Jun 2027 | 92   | 4.1304   | AUD 1,033,470 | AUD 1,041,096 |
| 23 Jun 2027 | 23 Sep 2027 | 92   | 4.1703   | AUD 1,033,470 | AUD 1,051,151 |

The fixed payments differ only through the number of days in each quarter. The floating payments follow the forward curve, rising from 4.02% for the first quarter to 4.17% for the last. So the fixed receiver collects more than it pays early on and pays more later; at the par rate the two legs are worth exactly the same.

## Where the par rate comes from

The fixed leg of a swap with fixed rate  $S$  is worth

$$PV_{\text{fixed}} = N \cdot S \sum_i \tau_i P(t_i) = N \cdot S \cdot A$$

where the sum  $A = \sum_i \tau_i P(t_i)$  is the **annuity**: the value today of receiving one unit per year, paid in the swap's periods. The floating leg is worth

$$PV_{\text{float}} = N \sum_i \tau_i F_i P(t_i)$$

where  $F_i$  is the forward rate for period  $i$ . With a single curve, each forward is a ratio of discount factors,

$$\tau_i F_i = \frac{P(t_{i-1})}{P(t_i)} - 1,$$

so the floating leg telescopes to  $N [P(t_0) - P(t_n)]$ . Setting the legs equal gives the **par swap rate**:

$$S = \frac{P(t_0) - P(t_n)}{A}$$

## Hand check: the three-year swap

For the three-year swap, which is one of our quotes, the curve gives  $P(t_0) = 0.999890$  at the start and  $P(t_n) = 0.881280$  at the end of its 12 quarterly periods, and an annuity of 2.810674.

|             | BY HAND  | QUANTLIB | QUOTE   |
|-------------|----------|----------|---------|
| Annuity $A$ | 2.810674 | 2.810674 |         |
| Par rate    | 4.2200%  | 4.2200%  | 4.2200% |

The hand calculation reproduces our three-year quote and QuantLib's fair rate exactly.

## An off-market swap

Now suppose we receive 4.50% for three years instead of the par rate of 4.22%. The floating leg is unchanged, so the whole value comes from the difference in fixed rates:

$$V_{\text{receiver}} = N (K - S) A$$

With  $K = 4.50\%$ , that is AUD 786,989 on AUD 100 million. QuantLib's value, AUD 786,989, agrees to within one ten-millionth of a dollar.

This is how traders think about swaps that are no longer at market: the annuity converts a rate difference into money.

## DV01 and PV01

The annuity is also the swap's risk. Bump every quote by one basis point, rebuild the curve and reprice the three-year swap at par:

| SHIFT                                  | VALUE             |
|----------------------------------------|-------------------|
| all quotes +1bp                        | AUD -28,102       |
| all quotes -1bp                        | AUD 28,111        |
| <b>DV01</b>                            | <b>AUD 28,107</b> |
| <b>PV01</b> = $N \cdot A \cdot 0.0001$ | <b>AUD 28,107</b> |

For a swap at par the two agree almost exactly: a one basis point change in the swap rate is worth the notional times the annuity times one basis point. Chapter 8 shows that this is not quite true once curves are built from several kinds of quote.

## Summary

- A vanilla AUD swap pays **quarterly against 3-month BBSW** out to three years and **semi-annually against 6-month BBSW** from four.
- A BBSW rate is **set at the start** of its period and paid at the end; an AONIA leg is known only at the end.
- The **par rate** is the floating leg divided by the **annuity**; with one curve, the floating leg is  $P(t_0) - P(t_n)$ .
- An off-market swap is worth  $N(K - S)A$ , and the swap's **PV01** is  $N \cdot A \cdot 1\text{bp}$ .

## Sources

Companion notebook: `ep04_bbsw_swaps.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. ASX, *OTC Clearing (eligible products: BBSW 3m, 6m vs Fixed; BBSW basis; AONIA)*. <https://www.asx.com.au/markets/clearing-and-settlement-services/otc-clearing>
3. Reserve Bank of Australia, *Fallbacks for BBSW Securities (Bulletin, June 2022)*. <https://www.rba.gov.au/publications/bulletin/2022/jun/fallbacks-for-bbsw-securities.html>

# Dual-Curve Pricing

*Australian dollar swaps pay BBSW but are discounted on the AONIA curve. We build both curves, compare the result with the single-curve set-up of Chapter 4, and find out where the difference shows up.*

## Why two curves

Most swaps today are cleared through a central counterparty or collateralised under a credit support agreement. Cash collateral earns an overnight rate, so the value of the swap's cash flows is discounted at the overnight rate too; clearing houses keep the interest they pay on collateral and the curve they discount with aligned. AFMA's conventions make this the standard for Australian dollar swaps: prices for uncleared derivatives are quoted as if discounted in the same way as cleared ones, which currently means **AUD OIS curves**.

BBSW is still what the floating leg pays. So we need two curves:

- the **AONIA curve** to discount every cash flow, and
- a **BBSW curve** only to forecast the floating payments.

## What changes in the maths

With one curve (Chapter 4), each forward was a ratio of the discounting curve's own factors, and the floating leg telescoped:

$$\sum_i \tau_i F_i P(t_i) = P(t_0) - P(t_n)$$

With two curves, the forwards come from the BBSW curve and the discount factors from AONIA, and nothing cancels:

$$PV_{\text{float}} = N \sum_i \tau_i F_i^{\text{BBSW}} P^{\text{AONIA}}(t_i)$$

The par rate is still the floating leg divided by the annuity, but both are now discounted on AONIA:

$$S = \frac{\sum_i \tau_i F_i^{\text{BBSW}} P^{\text{AONIA}}(t_i)}{\sum_i \tau_i P^{\text{AONIA}}(t_i)}$$

The shortcut really is gone. For our three-year swap at par, the floating leg is worth AUD 11,889,053, while  $N[P(t_0) - P(t_n)]$  on the AONIA curve would give AUD 11,459,117.

## Building the two curves

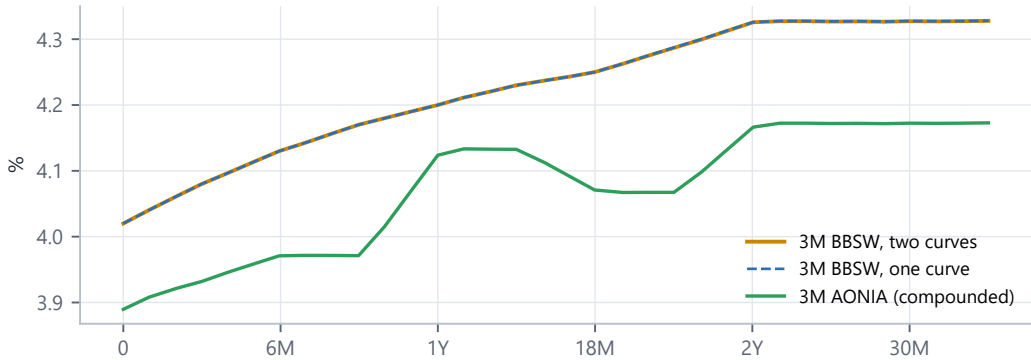
In QuantLib the change is a single argument. The AONIA curve is built exactly as in Chapter 3. When the BBSW swap helpers are created, they are given the AONIA curve as their **discounting curve**, so the bootstrap solves only for the BBSW forwards. The BBSW fixing and the FRAs don't involve discounting at all, so their helpers are unchanged.

The BBSW curve, with or without AONIA discounting.

```
def build_bbsw_curve(discount=None, bump_bp=0.0):
    index = bbsw3m()
    discounting = discount if discount is not None else ql.YieldTermStructureHandle() #
    empty = one curve
    helpers = []
    for row in bbsw_quotes.itertuples():
        q = ql.QuoteHandle(ql.SimpleQuote(row.rate_pct / 100 + bump_bp / 1e4))
        if row.instrument == "BBSW":
            h = ql.DepositRateHelper(q, index)
        elif row.instrument == "FRA":
            h = ql.FraRateHelper(q, int(row.tenor.split("x")[0]), index)
        else:
            h = ql.SwapRateHelper(q, ql.Period(row.tenor), cal, ql.Quarterly,
                                   ql.ModifiedFollowing, dc, index, ql.QuoteHandle(),
                                   ql.Period(0, ql.Days), discounting, 1)
        helpers.append(h)
    c = ql.PiecewiseLogLinearDiscount(today, helpers, dc)
    c.enableExtrapolation()
    return c

single = build_bbsw_curve() # forecasts and discounts
dual = build_bbsw_curve(discount=ois_handle) # forecasts only; AONIA discounts
single_h, dual_h = ql.YieldTermStructureHandle(single), ql.YieldTermStructureHandle(dual)
```

## Forwards and discount factors



**Figure 5.1.** Three-month forward rates: BBSW from the two-curve build (solid), from the single-curve build of Chapter 4 (dashed), and AONIA compounded over three months. Illustrative quotes.

The two BBSW forward curves are almost identical: they never differ by more than 0.06 of a basis point. The gap between BBSW and AONIA, between 8 and 22 basis points here, is what BBSW carries over the overnight rate for lending to banks for a term.

Discount factors change more. The AONIA curve sits below the BBSW curve, so its discount factors are higher: future cash flows are worth a little more.

Discount factors from the single BBSW curve and from the AONIA curve.

| FROM SPOT | BBSW CURVE | AONIA CURVE | DIFFERENCE |
|-----------|------------|-------------|------------|
| 1Y        | 0.959917   | 0.961437    | 0.001521   |
| 2Y        | 0.919931   | 0.922712    | 0.002781   |
| 3Y        | 0.881280   | 0.885303    | 0.004023   |

## A swap at par doesn't notice

Both builds were calibrated to the same three-year swap quote of 4.22%. So in both, the three-year par rate is 4.2200% and a swap at that rate is worth zero. The BBSW forwards shift, very slightly, to absorb the change in discounting.

## An off-market swap does

Receive 4.50% for three years again. Its value is still  $N(K - S)A$ , but the annuity is now discounted on AONIA:

|                | ONE CURVE   | TWO CURVES  |
|----------------|-------------|-------------|
| Annuity $A$    | 2.8107      | 2.8173      |
| Receiver value | AUD 786,989 | AUD 788,847 |

The difference is AUD 1,858 on this one swap. Across a book of older swaps, far from today's rates, such differences add up, which is why getting the discount curve right matters for valuation even though par rates don't depend on it.

## One cash flow by hand

The first floating payment of the two-curve swap runs 91 days from Wednesday 23 September 2026 to Wednesday 23 December 2026.

1. Its rate is the BBSW forward for the period, from the BBSW curve: 4.0207%.
2. On AUD 100 million that is a payment of AUD 1,002,411.57.
3. Discounted with the AONIA discount factor of 0.990290, it is worth AUD 992,678.50 today.

QuantLib gives AUD 992,678.50: the same, to the cent.

## Which curve carries the risk?

Bump one set of quotes by one basis point at a time, rebuild both curves (the BBSW curve depends on the AONIA curve through discounting) and reprice.

Change in value to the fixed receiver, three-year swaps on AUD 100 million.

| BUMP, THEN REBUILD BOTH CURVES | PAR SWAP    | OFF-MARKET SWAP |
|--------------------------------|-------------|-----------------|
| AONIA quotes +1bp              | AUD 0       | AUD -122        |
| BBSW quotes +1bp               | AUD -28,173 | AUD -28,173     |
| Both +1bp                      | AUD -28,169 | AUD -28,290     |

Move the AONIA quotes and hold the BBSW quotes fixed, and the par swap does not move at all: it still reprices to its quote. The off-market swap moves only a little, through its annuity. Move the BBSW quotes instead, and both swaps lose about AUD 28,200.

So with the BBSW quotes held fixed, a BBSW swap's rate risk sits almost entirely in those quotes. What you hold fixed when you bump decides how risk splits between curves. Chapter 6 introduces the quotes that make this choice natural: basis swaps.

## Summary

- **Forecast on BBSW, discount on AONIA:** the standard for AUD swaps.
- With two curves the floating leg no longer telescopes.
- **Par swaps** price the same on one curve or two; **off-market swaps** do not, because their value runs through the annuity.
- How risk splits between curves depends on **which quotes you hold fixed**.

---

## Sources

Companion notebook: `ep05_dual_curve.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. LCH Limited, *LCH Limited Self-Certification: RFR PAI and discounting transition (filed with the CFTC, June 2020)*. <https://www.cftc.gov/sites/default/files/filings/orgrules/20/06/rule061820lchltddco001.pdf>
3. Reserve Bank of Australia, *Fallbacks for BBSW Securities (Bulletin, June 2022)*. <https://www.rba.gov.au/publications/bulletin/2022/jun/fallbacks-for-bbsw-securities.html>

# Basis Swaps

---

*Short AUD swaps pay 3-month BBSW, longer ones 6-month BBSW, and all are discounted on AONIA. Basis swaps tie the three rates into one consistent family of curves.*

## Two floating legs

A **basis swap** exchanges one floating rate for another; there is no fixed leg. Two kinds matter for Australian dollar curves.

- **3s6s basis swaps**: one side pays 3-month BBSW every quarter, the other pays 6-month BBSW every six months.
- **AONIA/BBSW basis swaps** (AFMA calls them Cash/BBSW swaps, or "BOBs"): AONIA, compounded over each quarter, against 3-month BBSW.

The price of a basis swap is a **spread**, and AFMA's convention puts it on the **shorter leg**: 3-month BBSW in a 3s6s, AONIA in an AONIA/BBSW swap. In an AONIA/BBSW swap the spread is paid as simple interest at the end of each period. Like other single-currency AUD swaps, basis swaps start at T+1, and ASX clears both kinds.

## Why the spread isn't zero

BBSW includes a credit premium over a risk-free curve: it is what the prime banks pay to borrow for a term. That is why BBSW has usually sat above AONIA, the overnight rate. The same logic runs along the BBSW tenors. Lending to banks for six months carries more of that premium than lending for three, so six-month BBSW is normally above three-month, and the side paying three-month BBSW adds a spread to make the swap fair.

## Our quotes

Illustrative basis quotes, as spreads on the shorter leg. Not market data.

| TENOR | AONIA/BBSW (BP) | 3S6S (BP) |
|-------|-----------------|-----------|
| 1Y    | 13.0            | 8.0       |
| 2Y    | 14.0            | 9.0       |
| 3Y    | 15.0            | 10.0      |
| 5Y    | 15.5            | 11.0      |
| 7Y    | 16.0            | 11.5      |
| 10Y   | 16.5            | 12.0      |

Together with today's BBSW fixings (3-month 4.02%, 6-month 4.14%) and the AONIA swap quotes of Chapter 3, these are all the inputs we need.

## One curve family

The curves build in a chain.

1. **AONIA**, from the overnight index swaps, exactly as in Chapter 3. It discounts everything.
2. **3-month BBSW**: the 3-month fixing, then the AONIA/BBSW basis swaps. The AONIA leg of each basis swap is already known from step 1, so each quote pins down the BBSW forwards.
3. **6-month BBSW**: the 6-month fixing, then the 3s6s basis swaps, built on top of the finished 3-month curve.

QuantLib has a rate helper for each kind of basis swap. The overnight-to-BBSW helper takes the AONIA index, already linked to its curve, and solves for 3-month BBSW; the BBSW-to-BBSW helper takes the finished 3-month index and solves for 6-month. Both discount on AONIA.

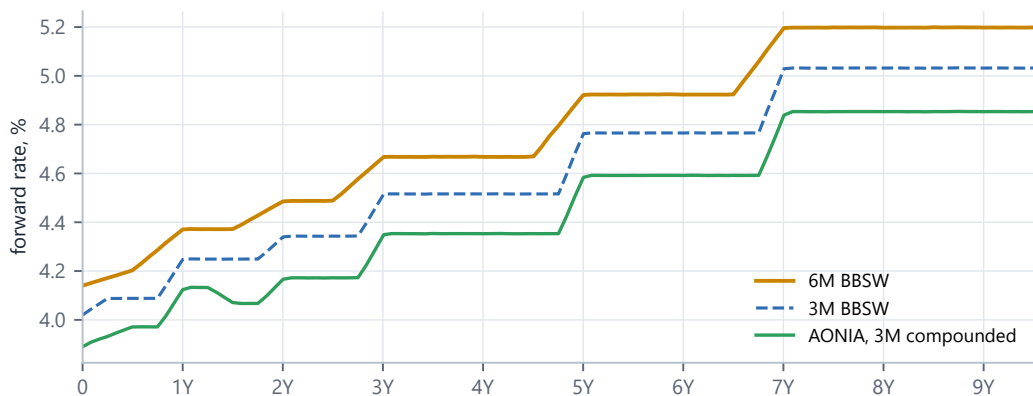
The 3-month and 6-month BBSW curves from basis swap helpers.

```
h3 = []
for r in rows("BBSW3M"):
    if r.instrument == "fixing":
        h3.append(ql.DepositRateHelper(handle(r), bbsw(3)))
    else: # AONIA + spread vs 3M BBSW, both quarterly
        h3.append(ql.OvernightIborBasisSwapRateHelper(
            handle(r), ql.Period(r.tenor), 1, cal, MF, False,
            aonia_index(aonia_h), bbsw(3), aonia_h))
bbsw3m = ql.PiecewiseLogLinearDiscount(today, h3, dc)
bbsw3m.enableExtrapolation()
bbsw3m_h = ql.YieldTermStructureHandle(bbsw3m)

h6 = []
for r in rows("BBSW6M"):
    if r.instrument == "fixing":
        h6.append(ql.DepositRateHelper(handle(r), bbsw(6)))
    else: # 3M BBSW + spread (quarterly) vs 6M BBSW (semi-annual)
        h6.append(ql.IborIborBasisSwapRateHelper(
            handle(r), ql.Period(r.tenor), 1, cal, MF, False,
            bbsw(3, bbsw3m_h), bbsw(6), aonia_h, False))
bbsw6m = ql.PiecewiseLogLinearDiscount(today, h6, dc)
bbsw6m.enableExtrapolation()

helpers = {"AONIA": ois_helpers, "BBSW3M": h3, "BBSW6M": h6}
curves = {"AONIA": aonia, "BBSW3M": bbsw3m, "BBSW6M": bbsw6m}
for c in curves.values():
```

Every quote reprices to within one hundred-millionth of a basis point.



**Figure 6.1.** The curve family: AONIA compounded over three months, 3-month BBSW, and 6-month BBSW, each over its own tenor starting at each date. Illustrative quotes.

The figure shows the structure: AONIA at the bottom, 3-month BBSW above it by the AONIA/BBSW spread, and 6-month BBSW above that by the 3s6s spread. The steps come from the log-linear interpolation of Chapter 3: flat forwards between pillars.

**A modelling note.** AFMA's conventions say AONIA/BBSW swap settlements are paid two business days after maturity. QuantLib's basis-swap helper has no payment lag, so the curves here ignore it. The effect on the curve is a small fraction of a basis point.

## Pricing a basis swap by hand

Take a five-year 3s6s swap: receive 6-month BBSW semi-annually, pay 3-month BBSW **plus a spread**  $s$  quarterly, both legs discounted on AONIA. At the par spread the legs are worth the same:

$$\underbrace{\sum_j N \tau_j F_j^{6M} P(t_j)}_{PV_{6M}} = \underbrace{\sum_i N \tau_i F_i^{3M} P(t_i)}_{PV_{3M}} + s \cdot N \underbrace{\sum_i \tau_i P(t_i)}_{A_{3M}}$$

so

$$s = \frac{PV_{6M} - PV_{3M}}{N \cdot A_{3M}}$$

On AUD 100 million, the 10 semi-annual payments of the 6-month leg are worth AUD 19,985,359 and the 20 quarterly payments of the 3-month leg AUD 19,490,131. The quarterly annuity is 4.502076.

|                 | BY HAND | QUANTLIB | QUOTE   |
|-----------------|---------|----------|---------|
| Par spread (bp) | 11.0000 | 11.0000  | 11.0000 |

## What the curves say swaps should be

With the family in place we can price the swaps people actually trade: quarterly against 3-month BBSW out to three years, and semi-annual against 6-month BBSW from four.

Par rates of vanilla AUD swaps, receive fixed, discounted on AONIA. Illustrative quotes.

| TENOR | FLOATING LEG         | PAR RATE (%) |
|-------|----------------------|--------------|
| 1Y    | 3M BBSW, quarterly   | 4.0710       |
| 2Y    | 3M BBSW, quarterly   | 4.1587       |
| 3Y    | 3M BBSW, quarterly   | 4.2174       |
| 4Y    | 6M BBSW, semi-annual | 4.4162       |
| 5Y    | 6M BBSW, semi-annual | 4.4623       |
| 7Y    | 6M BBSW, semi-annual | 4.5801       |
| 10Y   | 6M BBSW, semi-annual | 4.7370       |

The three-year rate is 4.22% and the four-year rate 4.42%. Part of that step is the extra year; part is the switch from 3-month to 6-month BBSW, which carries the 3s6s spread. In our curves that shows up as a jump between three and four years.

## Basis risk

Finally, the five-year 3s6s swap's risk. Bump one group of quotes by one basis point, let the curves rebuild, and reprice.

| QUOTES MOVED +1BP | CHANGE IN VALUE |
|-------------------|-----------------|
| AONIA swaps       | AUD 0           |
| AONIA/BBSW basis  | AUD 0           |
| 3s6s basis        | AUD 45,021      |

Move the AONIA quotes and the basis swap barely notices: both legs float, so they move together. Move the AONIA/BBSW spreads and it still barely moves, because both BBSW curves shift together. But move the 3s6s spreads and it changes by about AUD 45,000 per basis point. That is **basis risk**: exposure to the spread between two floating rates, with almost no outright rate risk.

AFMA's customary 3s6s parcel is sized to carry about AUD 40,000 of risk per basis point, which for this swap means a notional of about AUD 88.8 million.

## Summary

- Basis swaps exchange **one floating rate for another**, with the spread on the **shorter leg**.
- **AONIA/BBSW** spreads give the 3-month BBSW curve from AONIA; **3s6s** spreads give the 6-month curve from the 3-month.
- A basis swap's par spread is the difference between its legs divided by the notional times the annuity of the spread leg.
- Basis swaps carry **basis risk, not rate risk**.

## Sources

Companion notebook: `ep06_basis_swaps.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. ASX, *ASX Bank Bill Swap Rate (BBSW) Conventions and BBSW Methodology (effective 19 December 2025)*. <https://www.asx.com.au/content/dam/asx/benchmarks/asx-bbsw-conventions.pdf>
3. Reserve Bank of Australia, *Fallbacks for BBSW Securities (Bulletin, June 2022)*. <https://www.rba.gov.au/publications/bulletin/2022/jun/fallbacks-for-bbsw-securities.html>

# Valuing a Live Swap

*A swap is traded once but valued every day until it matures. We take a five-year swap traded in July 2025 and value it fourteen months later: what's already paid, what's already set, and what's still unknown.*

## The trade

| TERM                      | VALUE                           |
|---------------------------|---------------------------------|
| Side                      | receive fixed, pay 6-month BBSW |
| Notional                  | AUD 100 million                 |
| Fixed rate (illustrative) | 3.95%                           |
| Payments                  | semi-annual on both legs        |
| Traded                    | Monday 14 July 2025             |
| Start                     | Tuesday 15 July 2025            |
| Matures                   | Monday 15 July 2030             |
| Valued on                 | Tuesday 22 September 2026       |

A five-year swap pays semi-annually against 6-month BBSW, with the usual conventions: actual/365 fixed on both legs, a T+1 start, Modified Following with no end-of-month rule. On the valuation date, 2 of its 10 periods have been paid, the last on Wednesday 15 July 2026, and about 3.8 years remain.

Today's curves are the illustrative curve family of Chapter 6. The fixed rate and the past BBSW fixing below are also made up for teaching.

## What has changed since the trade

1. **Past payments are settled.** They are no longer part of the swap's value.
2. **The current period's rate is set.** The period that started on Wednesday 15 July 2026 had its 6-month BBSW rate fixed that day, at 4.20%. It is now history: it can't come from today's curve.

3. **Later periods are forecast** from today's 6-month BBSW curve.
4. **Everything is discounted** on today's AONIA curve.

## The past fixing

A pricing library has to be told the fixings that have already happened. Ask QuantLib for the swap's value before adding the fixing, and it stops with an error:

*Missing BBSW6M6M Actual/365 (Fixed) fixing for July 15th, 2026*

It will not invent a rate that has already been published. Adding the fixing to the BBSW index fixes that, and every instrument on the same index can then use it. Production systems work the same way, which is why a clean, complete history of fixings matters as much as today's quotes.

Without the fixing QuantLib refuses to price; with it, the swap values.

```
try:
    swap.NPV()
    missing_error = None
except RuntimeError as e:
    missing_error = str(e).splitlines()[0].split(": ", 1)[-1] # drop QuantLib's "2nd Leg: "
    prefix
    print("without the fixing:", missing_error)

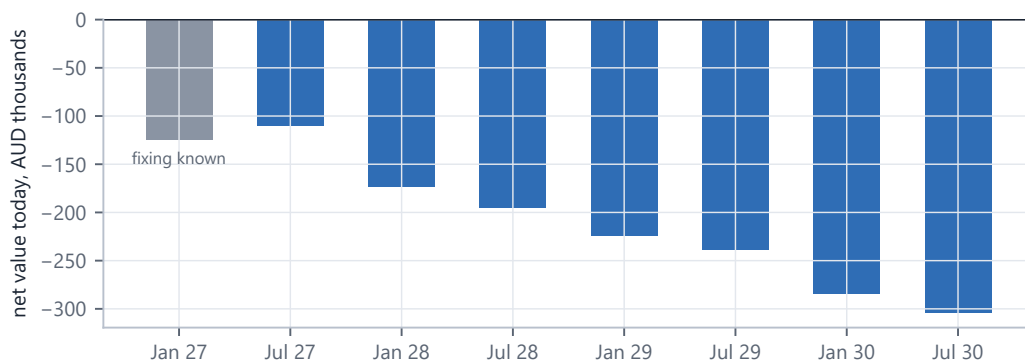
fixing_date = ql.DateParser.parseISO(current_fixing_date)
bbsw(6).addFixing(fixing_date, current_fixing_pct / 100) # fixings are stored by index name
out["fixing"] = {"date": current_fixing_date, "rate_pct": current_fixing_pct,
"missing_error": missing_error}
```

## What's left to pay

The remaining payments. Net value = (fixed – floating) × AONIA discount factor.

| PAID ON     | DAYS | BBSW (%) | KNOWN? | FIXED     | FLOATING  | NET VALUE TODAY |
|-------------|------|----------|--------|-----------|-----------|-----------------|
| 15 Jan 2027 | 184  | 4.2000   | yes    | AUD 1.99m | AUD 2.12m | AUD -124,497    |
| 15 Jul 2027 | 181  | 4.1794   | no     | AUD 1.96m | AUD 2.07m | AUD -110,226    |
| 17 Jan 2028 | 186  | 4.3089   | no     | AUD 2.01m | AUD 2.20m | AUD -173,561    |
| 17 Jul 2028 | 182  | 4.3716   | no     | AUD 1.97m | AUD 2.18m | AUD -195,497    |
| 15 Jan 2029 | 182  | 4.4430   | no     | AUD 1.97m | AUD 2.22m | AUD -223,959    |
| 16 Jul 2029 | 182  | 4.4876   | no     | AUD 1.97m | AUD 2.24m | AUD -239,229    |
| 15 Jan 2030 | 183  | 4.5992   | no     | AUD 1.98m | AUD 2.31m | AUD -284,326    |
| 15 Jul 2030 | 181  | 4.6676   | no     | AUD 1.96m | AUD 2.31m | AUD -304,239    |

The first remaining payment uses the known fixing; the rest use 6-month BBSW forwards, rising from 4.18% to 4.67%. Every one of them is above our fixed rate of 3.95%, so every net payment is against us.



**Figure 7.1.** Net value today of each remaining payment. The first (grey) uses the known fixing; the rest use forecast rates.

Adding them up, the swap is worth AUD -1,655,534 to us: the fixed leg is worth AUD 14,562,910 and the floating leg AUD -16,218,443.

## Two pieces you can check by hand

There is a neater way to see the value. Split the swap at the next reset date  $t_1$ , Friday 15 January 2027.

- **The current period.** Both rates are known, so it is a single net payment: the fixed rate  $K$  minus the fixing  $R_0$ , times the year fraction  $\tau_0$  (184 days), discounted from its payment date.
- **The rest** is a forward-starting swap from  $t_1$  to maturity, worth the notional times our rate minus today's par rate  $S_{\text{fwd}}$  for those dates, times their annuity  $A_{\text{fwd}}$  on AONIA.

$$V = \underbrace{N (K - R_0) \tau_0 P(t_1)}_{\text{current period}} + \underbrace{N (K - S_{\text{fwd}}) A_{\text{fwd}}}_{\text{forward-starting swap}}$$

Today's par rate for the forward-starting swap is 4.4301%, well above our 3.95%, and its annuity is 3.1888.

|                       | VALUE                    |
|-----------------------|--------------------------|
| Current period        | AUD -124,496.85          |
| Forward-starting swap | AUD -1,531,036.82        |
| <b>Total, by hand</b> | <b>AUD -1,655,533.67</b> |
| QuantLib              | AUD -1,655,533.67        |

The two pieces add up to QuantLib's value to the cent. The decomposition also tells the story of the trade: almost all of the loss comes from the fact that rates for the remaining life of the swap are now well above the rate we locked in.

## Dirty and clean value

Since the current period began, 69 days of interest have built up on both legs:

|                                          | AMOUNT                |
|------------------------------------------|-----------------------|
| Fixed accrued (we receive)               | AUD 746,712           |
| Floating accrued (we pay)                | AUD 793,973           |
| Net accrued                              | AUD -47,260           |
| <b>Dirty value</b> (what we computed)    | <b>AUD -1,655,534</b> |
| <b>Clean value</b> = dirty – net accrued | <b>AUD -1,608,273</b> |

The value computed above includes the accrued interest: it is the **dirty** value. Taking the net accrued out gives the **clean** value. The dirty value builds up with accrued interest and drops back when it is paid; the clean value removes that saw-tooth, which

makes it the better number for watching how the market has moved. QuantLib's accrued amounts, AUD 746,712 and AUD 793,973, match the hand calculation. Accrued interest here runs to the valuation date; some systems use the settlement date instead.

## Summary

- Once a swap is live, **past payments drop out**, the **current rate is history**, and only the future is forecast.
- A pricing library needs the **past fixings**; QuantLib refuses to guess them.
- Value = **the known current period + a forward-starting swap**, each checkable by hand.
- **Clean value = dirty value – accrued interest.**

---

## Sources

Companion notebook: `ep07_seasoned_swap.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>

# Risk I: DV01

*How much does a swap make or lose if rates move one basis point? There is more than one answer, and the differences matter. We measure DV01 three ways, see which curve carries the risk, and test how far a single number can be trusted.*

## The trade

We receive fixed at par on a five-year swap, semi-annual against 6-month BBSW, on AUD 100 million. With today's curve family (Chapter 6), the par rate is 4.4623% and the swap's annuity is 4.4787. At par it is worth zero; the question is what happens next.

## A basis point of what?

"DV01" means the change in value for a one basis point move in rates. But *which* rates? There are three common answers.

- **Par DV01** moves the **market quotes**: here, every AONIA swap quote, with the basis spreads held fixed. All three curves are rebuilt and the swap repriced.
- **Zero DV01** skips the quotes and moves the **zero rates** of the curves directly, by one basis point each.
- **PV01** is a shortcut from the **annuity**:  $N \cdot A \cdot 0.0001$ , the value of one basis point on the fixed rate.

We measure the first two by bumping down one basis point and up one, and taking half the difference:

$$DV01 = \frac{V(-1bp) - V(+1bp)}{2}$$

This central difference is more accurate than a single bump, and gives the value **gained when rates fall**, so a receiver has a positive DV01. Firms differ on the sign convention, and on whether they bump up, down or both; a risk report should always say which.

Regulators use a one basis point shift too. Under the Basel FRTB rules, the delta sensitivity for interest rate risk is the **PV01**, measured by moving the rate at each tenor by one basis point (Chapter 9).

## In QuantLib

Every quote is a `SimpleQuote`, so the par version is short: shift each AONIA quote with `setValue`, ask the swap for its value, and shift it back. The curves rebuild themselves. For the zero version, each curve is wrapped in a `ZeroSpreadedTermStructure` whose spread starts at zero and is bumped instead.

Par DV01: bump every AONIA quote, reprice, restore.

```
aonia_keys = [k for k in quote_handles if k[0] == "AONIA"]

def shift_quotes(bp):
    for k in aonia_keys:
        quote_handles[k].setValue(quote_handles[k].value() + bp / 1e4)

def npv_quotes_shifted(bp):
    shift_quotes(bp)
    v = swap.NPV()
    shift_quotes(-bp)
    return v

par_dv01 = (npv_quotes_shifted(-1) - npv_quotes_shifted(+1)) / 2
```

## Three answers

|            | PAR DV01   | ZERO DV01  | PV01       |
|------------|------------|------------|------------|
| AUD per bp | AUD 43,884 | AUD 45,670 | AUD 44,787 |

The three are close, but not the same. Zero DV01 is about 1.041 times par DV01. Moving every quoted swap rate by one basis point does not move every zero rate by exactly one basis point, and vice versa: quotes are par rates with their own compounding and payment frequencies, zero rates are continuously compounded, and the curves are rebuilt in between.

None of them is wrong. They answer different questions: the par version is expressed in the instruments you would hedge with; the zero version is closer in spirit to the tenor-by-tenor sensitivities of Chapter 9. The lesson is to always state which rates were moved.

## Which curve carries the risk?

Now move the zero rates of one curve at a time.

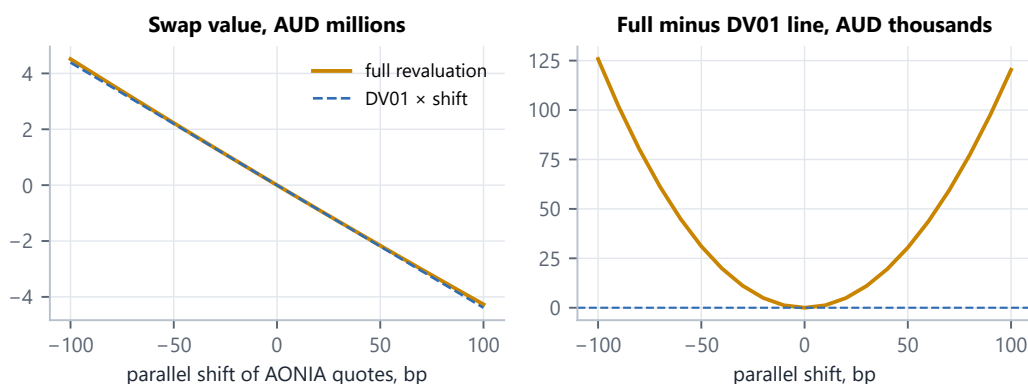
Central difference,  $\pm 1\text{bp}$  in continuously compounded zero rates.

| ZERO RATES MOVED             | DV01, AUD PER BP |
|------------------------------|------------------|
| 6-month BBSW zero rates only | AUD 45,787       |
| AONIA zero rates only        | AUD -117         |
| Both curves                  | AUD 45,670       |

Shift only the 6-month BBSW curve, the forecasting curve, and the swap moves by about AUD 45,800. Shift only the AONIA curve, which discounts, and it barely moves: about AUD 100. For a swap at par, a change in discounting changes the value of both legs almost equally, and they cancel. So the rate risk of a BBSW swap sits in its **forecasting curve**. (This is the zero-rate counterpart of the result in Chapter 5, where the risk sat in the BBSW quotes.)

## Bigger moves

A DV01 is a straight line: it predicts that a move of  $n$  basis points changes the value by  $n$  times DV01. Markets do not always move one basis point at a time. Figure 8.1 shifts every AONIA quote from  $-100$  to  $+100$  basis points in steps of ten, rebuilds the curves and fully revalues the swap each time.



**Figure 8.1.** Left: the swap's value under full revaluation and as predicted by DV01. Right: the difference between them.

On the left the two lines are almost indistinguishable. On the right, the gap between them is small near zero and grows on both sides.

| SHIFT  | DV01 PREDICTS  | FULL REVALUATION | DIFFERENCE  |
|--------|----------------|------------------|-------------|
| +100bp | AUD -4,388,389 | AUD -4,268,038   | AUD 120,351 |
| -100bp | AUD 4,388,389  | AUD 4,514,181    | AUD 125,792 |

At +100 basis points the real loss is smaller than DV01 predicts; at -100 the real gain is bigger. Either way the receiver comes out ahead of the straight line: by about AUD 120,400 on the way up and AUD 125,800 on the way down. That is **convexity**.

Where does it come from? The swap's value is roughly  $N(K - S)A$ , and the annuity  $A$  itself depends on rates: when rates rise, future payments are discounted more heavily and  $A$  shrinks, so each further basis point costs a little less; when rates fall,  $A$  grows and each basis point is worth a little more. For a plain swap the effect is small next to DV01, which is why DV01 is the number everyone quotes. For large moves, revalue rather than extrapolate.

## Summary

- DV01 is the value change for **one basis point**, but par, zero and annuity versions give **different numbers**: always say which rates were moved.
- The central difference  $[V(-1) - V(+1)]/2$  is the more accurate measure.
- A BBSW swap's rate risk sits in its **forecasting curve**; the discount curve's effect nearly cancels at par.
- **Convexity** makes a receiver do better than DV01 predicts in both directions; for big moves, revalue.

## Sources

Companion notebook: `ep08_dv01.ipynb`. Every number in this chapter is computed by it.

1. Basel Committee on Banking Supervision, *Basel Framework MAR21, Standardised approach: sensitivities-based method (MAR21.19, PV01 definition)*. <https://www.bis.org/committees/bcbs/basel-framework/standard/mar/21/inforce/2023-01-01/published/2020-03-27>

# Risk II: Bucketed Delta

*One DV01 number hides where along the curve the risk sits, and rates don't all move together. We split a four-year swap's risk by quote and by the Basel FRTB tenors, hedge it with the swaps the market trades, and find the basis risk that is left.*

## Two ways to bucket

There are two common ways to break a DV01 into pieces.

- **By quote** (par bucket delta): move one market quote at a time by one basis point, rebuild the curves, and reprice. Each bucket is the risk to one hedging instrument.
- **By tenor** (zero bucket delta): move the zero curve by one basis point at a fixed set of maturities, one at a time, tapering to zero at the neighbouring maturities.

Banks' regulatory capital under the Basel FRTB standardised approach uses the second kind. Its interest rate delta risk factors are defined per currency curve at ten tenors: 3 months, 6 months, and 1, 2, 3, 5, 10, 15, 20 and 30 years, and the sensitivity at each tenor is the PV01 for a one basis point move. The rules also say that an overnight index swap curve and an interbank offered rate curve must be treated as **different curves**, as must two such curves of different maturities. The Basel text gives euro examples; for Australian dollars the natural reading is AONIA, 3-month BBSW and 6-month BBSW as three curves, each with its own buckets.

This chapter shows sensitivities only. Capital also needs the regulatory risk weights and correlations, which are a topic of their own.

## The trade

We receive fixed at par, 4.4162%, on a **four-year** swap against 6-month BBSW, on AUD 100 million. Four years is deliberately awkward: our quotes have three-year and five-year points but nothing at four.

## Bucketing by quote

Par bucket delta of the four-year receiver. Buckets under AUD 1,000 not shown. Sign: change in value for +1bp.

| QUOTE MOVED +1BP    | VALUE CHANGE |
|---------------------|--------------|
| AONIA swaps 3Y      | AUD -13,441  |
| AONIA swaps 5Y      | AUD -22,398  |
| AONIA/BBSW basis 3Y | AUD -13,802  |
| AONIA/BBSW basis 5Y | AUD -22,966  |
| 3s6s basis 3Y       | AUD -13,802  |
| 3s6s basis 5Y       | AUD -22,966  |

Nobody quotes a four-year swap in our set, so the swap's rate risk lands on its neighbours: AUD -13,441 at the three-year AONIA quote and AUD -22,398 at the five-year, about 62% of it at five years, since four years is nearer to five than to three. The total over all AONIA quotes is AUD -35,848.

The basis quotes at three and five years show up too. That is because the BBSW curves are built on top of AONIA from the basis spreads (Chapter 6): a vanilla 6-month BBSW swap depends on AONIA swap quotes, AONIA/BBSW spreads and 3s6s spreads.

## Bucketing by FRTB tenor

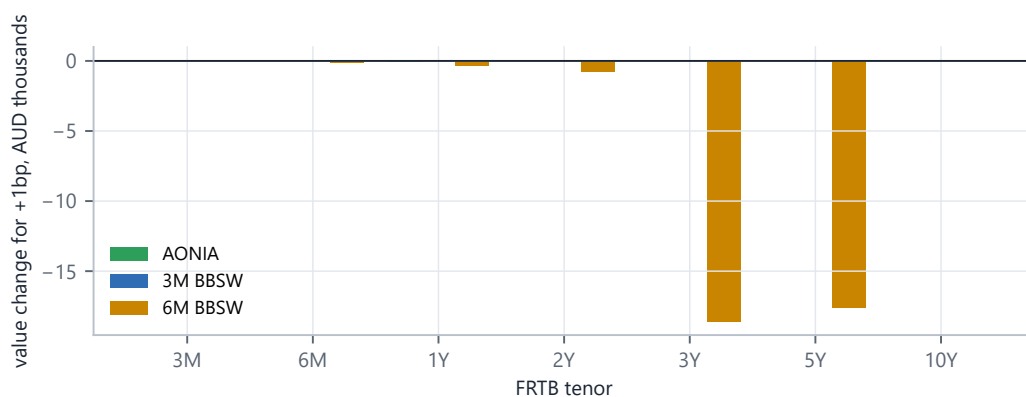
Each curve is wrapped in a piecewise zero spread, with one quote at each FRTB tenor, all set to zero. Setting one of them to one basis point moves that curve's zero rates by one basis point at that tenor, tapering linearly to zero at the neighbouring tenors: a **triangular bucket**.

Zero bucket deltas: bump one tenor of one curve at a time.

```
def zero_deltas(instr):
    table = {}
    for c in ("AONIA", "BBSW3M", "BBSW6M"):
        col = []
        for q in zq[c]:
            q.setValue(1e-4); up = instr.NPV()
            q.setValue(-1e-4); down = instr.NPV()
            q.setValue(0.0)
            col.append((up - down) / 2)
        table[c] = col
    return pd.DataFrame(table, index=FRTB_TENORS)
```

Zero bucket delta by FRTB tenor and curve: change in value for +1bp. The 15, 20 and 30-year buckets are zero.

| TENOR | AONIA   | 3M BBSW | 6M BBSW     |
|-------|---------|---------|-------------|
| 3M    | AUD 0   | AUD 0   | AUD 27      |
| 6M    | AUD -7  | AUD 0   | AUD -95     |
| 1Y    | AUD -12 | AUD 0   | AUD -338    |
| 2Y    | AUD -2  | AUD 0   | AUD -762    |
| 3Y    | AUD 63  | AUD 0   | AUD -18,627 |
| 5Y    | AUD 31  | AUD 0   | AUD -17,593 |
| 10Y   | AUD 0   | AUD 0   | AUD 0       |



**Figure 9.1.** Zero bucket delta of the four-year receiver by FRTB tenor, for each of the three curves.

Almost everything sits on the 6-month BBSW curve, the one that forecasts our payments: AUD -18,627 at three years and AUD -17,593 at five. The AONIA curve, which only discounts, barely registers, as Chapter 8 found for a swap at par. The 3-month curve is zero, because this swap never pays 3-month BBSW.

## Buckets add up

A useful check: the buckets on each curve should add up to moving the whole curve by one basis point.

| CURVE   | SUM OF BUCKETS | PARALLEL SHIFT |
|---------|----------------|----------------|
| 6M BBSW | AUD -37,387    | AUD -37,387    |
| AONIA   | AUD 74         | AUD 74         |

They match. Triangular buckets split the curve without gaps or overlaps: at any maturity, the neighbouring triangles add up to one.

## The hedge

To hedge, use the swaps the market actually trades. Out to three years that is quarterly against 3-month BBSW; from four years, semi-annual against 6-month BBSW. We pay fixed on a three-year quarterly swap and a five-year semi-annual swap, both at par, sized so that the portfolio has no risk to the three-year and five-year AONIA quotes: two equations in two unknown notionals.

Hedge notionals that net the 3-year and 5-year AONIA quote buckets to zero.

| HEDGE, AT PAR                        | NOTIONAL         |
|--------------------------------------|------------------|
| Pay fixed 3Y, quarterly vs 3M BBSW   | AUD 49.2 million |
| Pay fixed 5Y, semi-annual vs 6M BBSW | AUD 51.0 million |

## What's left

Sum of par bucket deltas in each group of quotes, before and after the hedge.

| QUOTES MOVED +1BP | 4-YEAR SWAP | HEDGES     | NET         |
|-------------------|-------------|------------|-------------|
| AONIA swaps       | AUD -35,848 | AUD 35,844 | AUD -4      |
| AONIA/BBSW basis  | AUD -36,767 | AUD 36,838 | AUD 70      |
| 3s6s basis        | AUD -36,767 | AUD 22,981 | AUD -13,787 |

The rate risk to the AONIA swap quotes is gone: what is left, AUD -4, is noise. The AONIA/BBSW basis risk nets out too, to AUD 70. But AUD -13,787 of **3s6s basis risk** remains, all of it at the three-year point. Our trade pays 6-month BBSW; the three-year hedge pays 3-month BBSW. Nothing in the hedge covers the difference between them.

## Closing the basis

That is exactly what the basis swaps of Chapter 6 are for. A three-year 3s6s basis swap, **receive 6M, pay 3M + spread**, on about AUD 48.9 million, cancels the remaining 3s6s risk. Rate risk is hedged with swaps; basis risk with a basis swap.

## Summary

- Bucket risk **by quote** (par) or **by tenor** (zero); the Basel FRTB rules use ten tenors per curve and treat AONIA, 3-month and 6-month BBSW as separate curves.
- Risk between two quoted tenors **splits between them**.
- Each curve's buckets **add up to its parallel shift**.
- Standard swaps hedge the rate risk of an off-tenor swap; the **basis risk** left over needs a **basis swap**.

This completes the path of the book: from one overnight rate, through the curves built on it, to a hedged book of AUD swaps.

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## Sources

Companion notebook: `ep09_bucketed_delta.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Basel Committee on Banking Supervision, *Basel Framework MAR21, Standardised approach: sensitivities-based method (MAR21.8 GIRR risk factors; MAR21.19 PV01)*. <https://www.bis.org/committees/bcbs/basel-framework/standard/mar/21/inforce/2023-01-01/published/2020-03-27>

# Appendix: AUD conventions at a glance

These are the conventions used throughout the book, with the source for each. Section numbers refer to AFMA's *Interest Rate Derivative Conventions* (June 2025) unless stated otherwise. Always check the current documents and your clearing house's product terms before relying on them.

## Rates

| RATE              | CONVENTION                                                                                                                                   | SOURCE                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Cash rate target  | set by the RBA Monetary Policy Board, eight meetings a year; announced 2:30pm Sydney time, effective the next day                            | RBA, <i>Cash Rate Target Overview</i>         |
| Cash rate (AONIA) | unsecured overnight interbank rate, calculated and published by the RBA; the rate for a day is published the next business day before 9:20am | RBA, <i>Cash Rate Methodology</i> ; AFMA §2.2 |
| BBSW              | 1 to 6-month prime bank paper; trades 8:30–10:00am; volume-weighted average, with fall-backs; published by 10:30am, four decimal places      | ASX, <i>BBSW Conventions and Methodology</i>  |

## Dates

| ITEM                   | CONVENTION                                               | SOURCE     |
|------------------------|----------------------------------------------------------|------------|
| Business day           | not a bank close day in New South Wales                  | §3.3       |
| Start of AUD swaps     | next business day after the trade (T+1)                  | §5.2, §3.3 |
| Date rolling           | Modified Following                                       | §3.3, §5.2 |
| End-of-month rule      | cross-currency swaps only, not single-currency AUD swaps | §3.3       |
| Forward-starting swaps | start and roll dates counted from the trade's spot date  | §3.3       |

## AONIA overnight index swaps

| ITEM                  | CONVENTION                                                           | SOURCE        |
|-----------------------|----------------------------------------------------------------------|---------------|
| Day count             | actual/365 fixed, both legs; no leap-year adjustment                 | \$3.7         |
| Floating amount       | $N[\prod(1 + r_i n_i / 365) - 1]$ , rounded to the cent              | \$3.17        |
| Weekends and holidays | counted in the day count of the previous business day                | \$3.17        |
| Last reset            | one Sydney business day before the end date                          | \$3.17        |
| Payments              | up to 12 months: once at maturity; longer: front stub, then annually | \$3.7, \$5.2  |
| Payment lag           | two business days after each period ends                             | \$3.17, \$5.2 |
| Customary size        | about AUD 25,000 of risk per basis point                             | \$3.4         |

## BBSW swaps and basis swaps

| ITEM                         | CONVENTION                                                                                              | SOURCE               |
|------------------------------|---------------------------------------------------------------------------------------------------------|----------------------|
| Vanilla swaps                | quarterly vs 3-month BBSW to 3 years; semi-annual vs 6-month BBSW from 4 years; 3 to 4 years negotiated | \$3.7, \$2.3         |
| Day count                    | actual/365 fixed, both legs                                                                             | \$3.7                |
| Floating rate                | set at the start of each period, paid in arrears                                                        | \$2.2                |
| 3s6s basis swaps             | 3-month BBSW quarterly vs 6-month BBSW semi-annually                                                    | \$2.3                |
| AONIA/BBSW (Cash/BBSW) swaps | AONIA compounded quarterly vs 3-month BBSW; paid two business days after maturity                       | \$2.3, \$3.17, \$5.2 |
| Basis spread                 | quoted on the shorter leg                                                                               | \$3.6                |
| Customary size, 3s6s         | about AUD 40,000 of risk per basis point                                                                | \$3.4                |
| Discounting                  | as for cleared derivatives: AUD OIS curves, cleared or not                                              | \$3.6                |

## Risk

| ITEM                            | DEFINITION                                                                                      | SOURCE                   |
|---------------------------------|-------------------------------------------------------------------------------------------------|--------------------------|
| FRTB interest rate delta tenors | 0.25, 0.5, 1, 2, 3, 5, 10, 15, 20 and 30 years, per currency curve                              | Basel Framework MAR21.8  |
| Separate curves                 | an OIS curve and an IBOR curve are different curves, as are IBOR curves of different maturities | Basel Framework MAR21.8  |
| Sensitivity                     | PV01: change in value for a one basis point move at each tenor                                  | Basel Framework MAR21.19 |

## Glossary

**Annuity.** The value today of receiving one unit per year in a swap's payment periods:  $A = \sum_i \tau_i P(t_i)$ . Converts a rate difference into money.

**Basis swap.** A swap of two floating rates, such as 3-month against 6-month BBSW.

**Bootstrapping.** Solving for a curve's discount factors one quote at a time, shortest maturity first.

**Clean value.** A swap's value excluding accrued interest.

**Convexity.** The change in DV01 as rates move; the gap between full revaluation and the straight-line DV01 prediction.

**Dirty value.** A swap's value including accrued interest.

**Discount factor.**  $P(t)$ : the value today of one unit paid at time  $t$ .

**DV01.** The change in value for a one basis point move in rates. Par DV01 moves market quotes; zero DV01 moves zero rates.

**FRA.** Forward rate agreement: locks in a BBSW rate for a future period.

**Front stub.** A shorter first period in a swap longer than twelve months.

**Par rate.** The fixed rate at which a swap is worth zero: the floating leg divided by the annuity.

**PV01.** The value of one basis point: for a swap,  $N \cdot A \cdot 0.0001$ ; in the Basel rules, the sensitivity at one tenor.

**Rate helper.** QuantLib's object for one quote in a bootstrap.

**Reset.** The date on which a floating rate is set.